FUNCTIONS

WHAT IS A FUNCTION?

- A function (called a method in C#) is a named block of code that performs a specific task.

Instead of writing the same code repeatedly, a function allows you to write it once and reuse it whenever needed.

WHY DO WE USE FUNCTIONS?

Functions are used to:

- ✅ Reduce code repetition
- ✅ Make programs easier to read and understand
- ✅ Organize code into logical parts
- ✅ Make programs easier to debug and maintain
- ✅ Encourage modular programming

TYPES OF FUNCTIONS IN C#

Built-in Functions

1. Provided by C# (e.g., Console.WriteLine())

User-Defined Functions

1. Created by the programmer to perform custom tasks

TYPES OF FUNCTIONS IN C#

Built-in Functions

1. Console.Write/WriteLine
2. Console.ReadLine
3. Math.Sqrt
4. Math.Pow
5. ToUpper
6. Length

FUNCTION SYNTAX

```csharp
access_modifier return_type MethodName(parameters)
{
    // Local variable declarations
    // Statements
    return value;
}
```

```csharp
static int Add(int a, int b)
{
    int sum = a + b;   // local variable
    return sum;
}
```

FUNCTION SYNTAX

Key Points:

- Methods must be inside a class
- static methods can be called without creating an object
- Local variables exist only within the method

CALLING A FUNCTION

```csharp
int result = Add(5, 3);
Console.WriteLine(result);
```

Key Points:

- Arguments must match parameter types
- Return values can be stored or used directly
- Methods improve code reusability

FUNCTION SCOPE

Types of Scope in C#:

- Local variables – declared inside methods
- Class-level variables (fields) – declared inside a class but outside methods

```csharp
class Sample
{
    int globalValue = 10; // class-level variable

    static void Display()
    {
        int localValue = 5; // local variable
        Console.WriteLine(localValue);
    }
}
```

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VARIABLE PARAMETERS

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VARIABLE PARAMETERS

Value Parameters (Default)

```csharp
static void ShowNumber(int x)
{
    Console.WriteLine(x);
}
```

Ref Parameters

```csharp
static void Increment(ref int x)
{
    x++;
}
```

VARIABLE PARAMETERS

out parameter

```csharp
static void GetValues(out int a, out int b)
{
    a = 10;
    b = 20;
}
```

THANK YOU